Supply Chain & Logistics Analytics

Data Science Business Analysis Report

A portfolio-level analytics solution covering sales, profitability, logistics, shipping, products, categories, markets, and regional performance.

Key Metric	Full Dataset Result
Total Sales	$36,784,735
Total Profit	$4,418,272
Overall Profit Margin	12.01%
Total Records / Orders	180,519
Units Sold	384,079
Average Order Value	$203.77
Average Delivery Delay	0.57 days
Delayed Orders	57.28%

Dashboard stack: Python • Pandas • NumPy • Plotly • Streamlit • Jupyter Notebook

1. Executive Summary

The Supply Chain & Logistics Analytics project provides an integrated view of commercial and operational performance. The final Streamlit dashboard combines executive KPIs, business insights, sales and revenue analysis, profitability, logistics and delivery performance, shipping-mode analysis, product/category analysis, regional analysis, and filtered-data exploration.

The dashboard is filter-driven: users can select Market, Order Region, Category, Shipping Mode, and Delivery Performance, after which the KPIs and analytical views recalculate for the selected subset.

2. Dataset & Data Preparation

- The analyzed dataset contains 180,519 records.
- The dashboard uses business fields including Sales, Profit, Product Name, Category Name, Market, Order Region, Shipping Mode, and delivery-performance measures.
- The analytical workflow validates numeric values, aggregates data at product, category, market, regional, and shipping levels, and derives business KPIs.
- Derived metrics include profit margin, average order value, average delivery delay, delayed-order percentage, order volume, and grouped sales/profit measures.

3. Executive Overview & Business Insights

The dashboard begins with two KPI rows and a dynamic Business Insights & Recommendations section. This placement gives users the summary and interpretation before moving into detailed charts.

- Total Sales: approximately $36.78 million.
- Total Profit: approximately $4.42 million.
- Overall Profit Margin: 12.01%.
- Units Sold: 384,079.
- Average Order Value: $203.77.
- Average Delivery Delay: 0.57 days.
- Delayed Orders: 57.28% under the full-dataset view.

4. Sales & Revenue Analysis

- Sales by Category identifies revenue concentration across product categories.
- Sales and profit by category provides a combined commercial view.
- Sales-versus-profit analysis highlights the relationship between revenue generation and profitability.
- Market and regional performance views extend the revenue analysis geographically.

5. Profitability Analysis

- Profit by category ranks the strongest category-level contributors.
- Category profit-margin analysis evaluates efficiency rather than sales volume alone.
- Top Products by Profit identifies high-value product contributors.
- Loss-Making Products Analysis isolates products with negative total profit for further review.
- Profit Margin by Shipping Mode compares profitability across shipping methods.
- Category Sales vs Profit and Top Categories by Profit Margin provide additional profitability perspectives.

6. Logistics & Delivery Performance

- Delivery Performance Analysis summarizes the distribution of delivery outcomes.
- Order Volume by Region shows where order activity is concentrated.
- Delivery Delay Analysis compares average delivery delay by shipping mode.
- Delivery Performance by Region compares delivery outcomes geographically.
- Regional delivery analysis supports identification of locations requiring operational attention.

7. Shipping Mode Analysis

- Shipping Mode Analysis evaluates operational and commercial differences between shipping methods.
- Sales & Profit by Shipping Mode compares revenue and profit contribution.
- Profit Margin by Shipping Mode compares profitability efficiency across shipping methods.
- Delivery Delay Analysis adds a logistics-efficiency perspective to the shipping comparison.

8. Product & Category Performance

- Product-level performance is evaluated using sales and profit rankings.
- Top Products by Profit highlights the strongest profit contributors.
- Loss-making products identify potential portfolio, pricing, cost, or fulfillment concerns.
- Category Sales vs Profit compares category scale with profitability.
- Top Categories by Profit Margin identifies categories with stronger percentage profitability.

9. Regional & Geographic Performance

- Regional sales and profitability views identify geographic revenue and profit concentration.
- Regional Profitability Analysis provides a direct comparison of regional profit contribution.
- Regional Sales vs Profit highlights regions combining scale and profitability.
- Order Volume by Region adds an operational measure of geographic demand.
- Market Performance Analysis provides a broader market-level comparison.

10. Interactive Filtering & Data Explorer

The dashboard includes sidebar filters for Market, Order Region, Category, Shipping Mode, and Delivery Performance. A Filter Status area reports the number of records remaining after the selected filters.

The Filtered Data Explorer presents the currently selected records in an interactive table, while the Download Filtered Data feature exports the filtered dataset as a CSV file.

11. Key Business Insights

- Sales volume should be evaluated together with profit and profit margin because high revenue does not automatically indicate strong profitability.
- The 12.01% overall profit margin provides a baseline for comparing categories, products, regions, markets, and shipping modes.
- Delivery performance is a major operational consideration because delayed orders represent a substantial share of the full dataset.
- Product-level analysis can distinguish high-value products from products generating negative profit.
- Regional and market comparisons can support targeted decisions rather than applying a single strategy across all locations.
- Shipping-mode analysis connects commercial contribution with delivery efficiency and profitability.

12. Business Recommendations

- Prioritize products and categories that combine strong sales with healthy profit margins.
- Review loss-making products for pricing, discounting, procurement cost, fulfillment cost, and demand factors.
- Investigate regions and shipping modes with weaker delivery performance or higher delays.
- Use profit margin, not sales alone, when evaluating products, categories, regions, and shipping modes.
- Monitor delivery performance continuously and investigate recurring delay patterns.
- Use dashboard filters to compare performance under specific market, regional, category, shipping, and delivery conditions before making decisions.

13. Dashboard Architecture & Project Flow

Stage	Dashboard Component
1	Page configuration, project paths, and data loading
2	Sidebar filters and filtered-record status
3	Executive Overview KPI cards
4	Business Insights & Recommendations
5	Sales & Revenue Analysis
6	Profitability and Sales vs Profit Analysis
7	Logistics, Shipping, Product, and Category Analysis
8	Regional, Market, and Delivery Analysis
9	Filtered Data Explorer and CSV export
10	Dashboard footer / project information

14. Technology Stack

- Python — core programming and analytics.
- Pandas — data cleaning, transformation, grouping, and aggregation.
- NumPy — numerical calculations and validation.
- Plotly — interactive analytical visualizations.
- Streamlit — interactive dashboard development.
- Jupyter Notebook — exploratory data analysis and preprocessing workflow.

15. Conclusion

The completed Supply Chain & Logistics Analytics dashboard brings together the full analytical workflow from filtered KPIs and business interpretation to detailed sales, profitability, logistics, shipping, product, category, market, and regional analysis. The final dashboard also provides filtered-data exploration and CSV export, making the solution useful for both analytical review and business decision support.

The report is intended to document the final dashboard and analytical project structure for portfolio and GitHub presentation.